Drying Chamber

Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Drying chambers or controlled rooms

What it is used for

A drying chamber or controlled drying room removes moisture from materials by maintaining specified temperature, airflow, and sometimes humidity conditions. Correct loading, monitoring, and process control help achieve uniform drying while protecting product quality and preventing overheating or contamination.

Main parts

- Insulated chamber or drying-room enclosure
- Heating system and temperature controller
- Air circulation fans and ducts
- Temperature and humidity sensors, where fitted
- Shelves, racks, trays, or trolleys
- Door seals, ventilation outlets, and control panel

Before you start

1. Confirm the material type, batch size, target moisture level, and approved drying temperature and duration.
2. Check that the chamber is clean, dry, and suitable for the product; verify that previous residues or contaminants have been removed.
3. Inspect heaters, fans, ducts, door seals, sensors, electrical connections, and safety controls for damage.
4. Verify that temperature and humidity indicators are functioning and calibrated where required.
5. Arrange material evenly on clean trays or racks, leaving enough space for airflow and avoiding overloading.
6. Confirm that the ventilation route is clear and that the material is safe to heat under the selected process conditions.

How to use

1. Switch on the chamber and set the temperature, airflow, humidity, and timer according to the approved process specification.
2. Allow the chamber to reach the required operating conditions and verify the displayed readings before loading, where the procedure requires preheating.
3. Load the prepared trays or racks carefully and ensure that air can circulate around the material.
4. Close the door securely and confirm that door seals and safety interlocks are working.
5. Start the drying cycle and record the batch identification, start time, setpoints, and initial readings.

6. Monitor temperature, humidity where applicable, airflow, and product condition at the specified intervals.
7. Rotate or reposition trays only if the approved procedure requires it and it can be done safely.
8. Adjust settings only within the authorised process limits if drying is uneven or readings deviate from the specification.
9. Stop the process if there is smoke, unusual odour, overheating, fan failure, electrical fault, or unexpected temperature rise.
10. At the end of the cycle, switch off heating and follow the procedure for cooling and ventilation before opening the chamber.
11. Remove the material using suitable gloves or handling tools if trays or products remain hot.
12. Check final moisture content or other specified quality parameters and record the results before releasing the batch.
13. Switch off the chamber when required, remove residues after cooling, and clean the chamber and trays using the approved method.

⚠ Safety

- Only trained personnel should operate the drying chamber and set process parameters.
- Do not exceed the chamber's rated temperature or use it for materials not approved for the equipment.
- Keep air inlets, outlets, and circulation fans unobstructed.
- Use suitable heat-resistant gloves and other required PPE when handling hot trays or materials.
- Do not bypass door interlocks, over-temperature cut-outs, or other safety controls.
- Isolate power before maintenance, cleaning near electrical components, or servicing heaters and fans.
- For flammable, reactive, or solvent-containing materials, use only equipment and procedures specifically approved for those hazards.

Care & storage

- Clean trays, racks, chamber surfaces, and air passages at the approved frequency.
- Inspect door seals, fans, heaters, sensors, and control wiring for wear or damage.
- Verify temperature and humidity sensors periodically using suitable calibrated instruments.
- Keep ventilation paths clear and remove dust or residue that could obstruct airflow.
- Arrange servicing if temperature control becomes unstable, airflow weakens, or safety cut-outs fail.

Common mistakes to avoid

- Overloading trays and restricting airflow through the chamber.
- Using incorrect temperature or drying time for the material.
- Ignoring sensor readings, uneven drying, or signs of overheating.
- Opening the chamber before following the required cooling and safe-handling procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the

manufacturer's original manual / file.